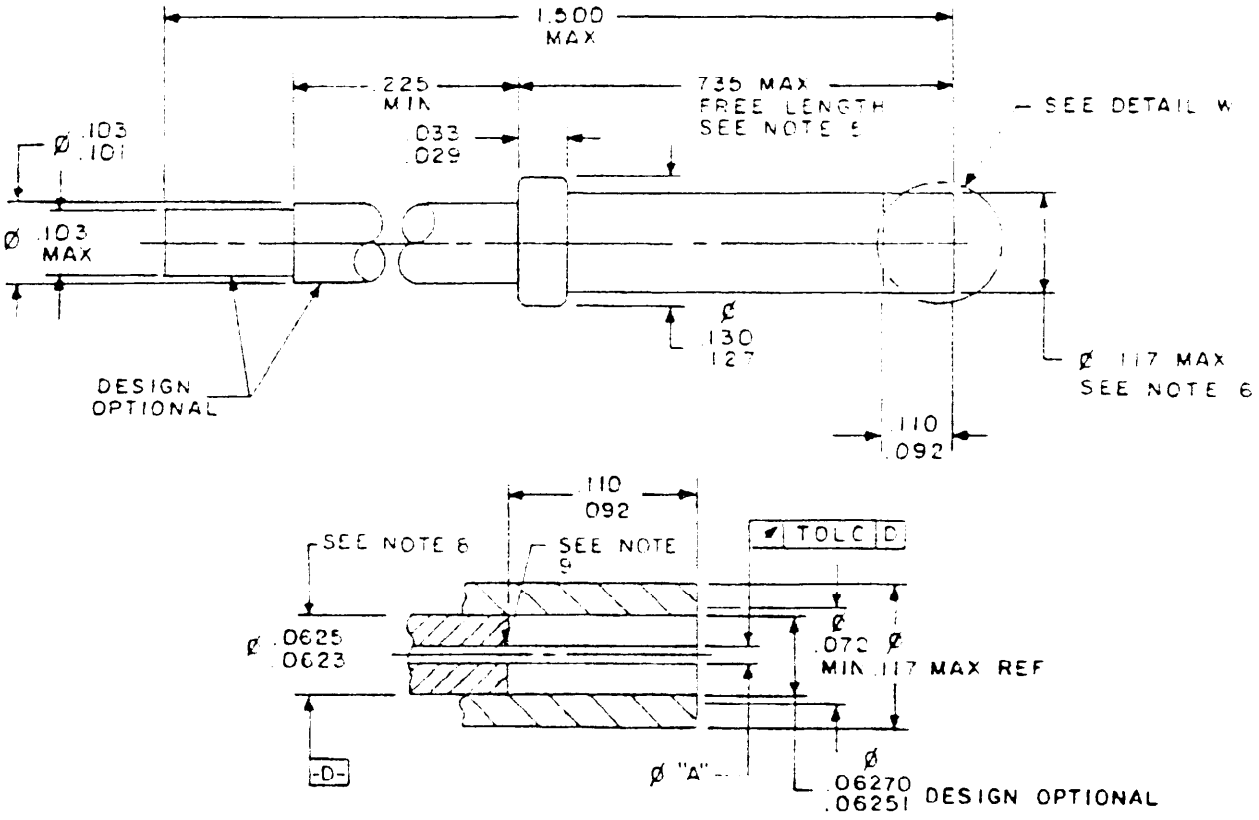
MILITARY SPECIFICATION SHEET

TERMINI, FIBER OPTIC, CONNECTOR, REMOVABLE, ENVIRONMENT RESISTING, CLASS S, TYPE II, STYLE A, SOCKET TERMINUS, SIZE 16, REAR RELEASE MIL-C-38999, SERIES I, III, and IV

This specification sheet is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-T-29504.



DETAIL W

FIGURE 1. Socket terminus.

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

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Inches	mm		Inches	mm
.029	0.74		.103	2.62
.033	0.84		.110	2.79
.072	1.83		.117	2.97
.0625	1.58		.127	3.23
.0627	1.59		.130	3.30
.084	2.13		.225	5.72
.092	2.34		.431	10.95
.100	2.54		.575	14.61
.101	2.57		.702	17.83
			1.500	38.10

NOTES:
1. Dimensions are in inches, except for diameter A which is dimensioned in metric microns only.
2. Metric equivalents are given for general information only.
3. All diameters to be concentric within .003 unless otherwise indicated.
4. Dimensions apply after plating when applicable.
5. Free length prior to installation in the connector.
6. Alignment sleeve and body design optional, not to exceed maximum envelope dimensions.
7. For alternate alignment sleeve design, engaging force shall be less than preload force (I.D. tolerance does not apply).
8. Alignment surface O.D. shall meet the circular runout as specified in table I.
9. Polished fiber face may be flat to concave and flush to recess.
10. "A" diameter hole tolerance is +B -0, see table I.

FIGURE 1. Socket terminus - Continued.

REQUIREMENTS:

Design and construction:

Dimensions and configuration: See figure 1 and table I.

Weight: 1 gram maximum.

Adhesives: Use a B-stage epoxy with a curing time of 15 minutes at 200°C. Epotec 354, or Epotec 353ND, or an equivalent epoxy approved by the preparing activity shall be used. Mixing and curing instructions shall be supplied with the assembly instructions provided with each termini or applicable connector.

Tools: See table II.

Mating terminus: MIL-T-29504/4.

Part or Identifying Number (PIN): See figure 1 and table I. The PIN M29504/5-(dash number from table I.)

Marking: The PIN shall be marked on barrel of the termini.

Terminus engagement and separation force: Maximum engagement force shall be 30 ounces.

Temperature life: The upper temperature limits shall be 150°C ±3°C.

Thermal shock: The temperature range used shall be -65°C to +150°C.

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TABLE I. "A" diameter hole for optical waveguide fiber and PIN designation.

TIOC	PIN	"A" diameter 1/ μ	Total allowable offset tolerance 2/		Fiber size	Terminus
			Circular runout 3/	Max $\frac{B + C}{2}$ (inch)μ	Core/clad μ	Size
4045	M29504/05-4045	124.5 +B -0	C			16
4046	M29504/05-4046	127.0 +B -0	C	(.00059) 15	50/125	16
4047	M29504/05-4047	129.0 +B -0	C			16
4048	M29504/05-4048	139.5 +B -0	C			16
4049	M29504/05-4049	142.0 +B -0	C	(.00059) 15	100/140	16
4050	M29504/05-4050	144.0 +B -0	C			16

4/ Dimensions in parentheses are inches.
3/ As defined by ANSI Y14.5M - 1982.
1/ B is defined as the clearance between the fiber and the hole. B and C are not specified, but calculated or selected by design.
2/ The total allowable offset tolerance is the tolerance that controls the offset of the fiber to the alignment diameter of .0625/.0623.

TABLE II. Tools.

Enc size	Insertion tool	Removal tool
16	M81969/14-03	M81969/14-03

Patent notice. The Government does not have a royalty-free license under the following listed patents for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

Patent number	Patent expiration date
3,947,182	3/30/1993
4,351,586	9/28/1997
4,747,658	5/31/2005

Revision letters are not used to denote changes due to the extensiveness of the changes.

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CONCLUDING MATERIAL

Custodians:
Army - CR
Navy - EC
Air Force - 85

Review activities:
Army - MI
Navy - AS, MC, SH
Air Force - 17, 19, 70, 80, 90, 99
DLA - ES

User activities:
Army - AV
Navy - CG, MC, OS, YD
Air Force - 13, 14, 84

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 6060-0092-2)

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